PREDICTIVE MAINTENANCE AGENT

Leveraging Machine Learning & IoT Sensor Data for Proactive Industrial Equipment Management

AGENDA

01 Problem Statement

Why industries need predictive maintenance

02 System Architecture

ML pipeline, IoT sensors & data flow

03 Key Features

Real-time monitoring, anomaly detection & alerts

04 Live Dashboard

Web interface & visualization

05 Results & Impact

KPIs, cost savings & future scope

SYSTEM ARCHITECTURE

IoT Sensors
Temp · Vib
Pressure · RPM

Data Ingestion
Stream · Buffer
Validate

Feature Engine
Normalize · Extract
Window

Prediction Engine
Risk Score
Anomaly Flag

Dashboard & Alerts
Web UI
SMS · Email

SENSORS & EDGE

- Temperature Sensors
- Vibration Accelerometers
- Pressure Transducers
- RPM Encoders

DATA PROCESSING

- Apache Kafka Streams
- Python Pandas Pipeline
- Sliding Window (30s)
- Anomaly Scoring

ML MODELS

- Random Forest Classifier
- LSTM Time-Series Model
- XGBoost for Risk Score
- Ensemble Voting

INTERFACE

- HTML / CSS / JS Dashboard
- Canvas-based Charts
- REST API (Flask)
- Email & SMS Alerts

KEY FEATURES

Real-Time Monitoring

Continuous 24/7 sensor data collection with 2-second refresh intervals across all assets.

Anomaly Detection

ML-driven outlier detection flags deviations beyond ±2σ from baseline behavior.

AI Prediction Engine

Ensemble model predicts failure probability up to 72 hours ahead with 94.7% accuracy.

Auto Work Orders

Automatically schedules maintenance tasks and dispatches technician alerts on critical findings.

Live Analytics

Interactive dashboard with rolling sensor charts, risk gauges, and KPI stat cards.

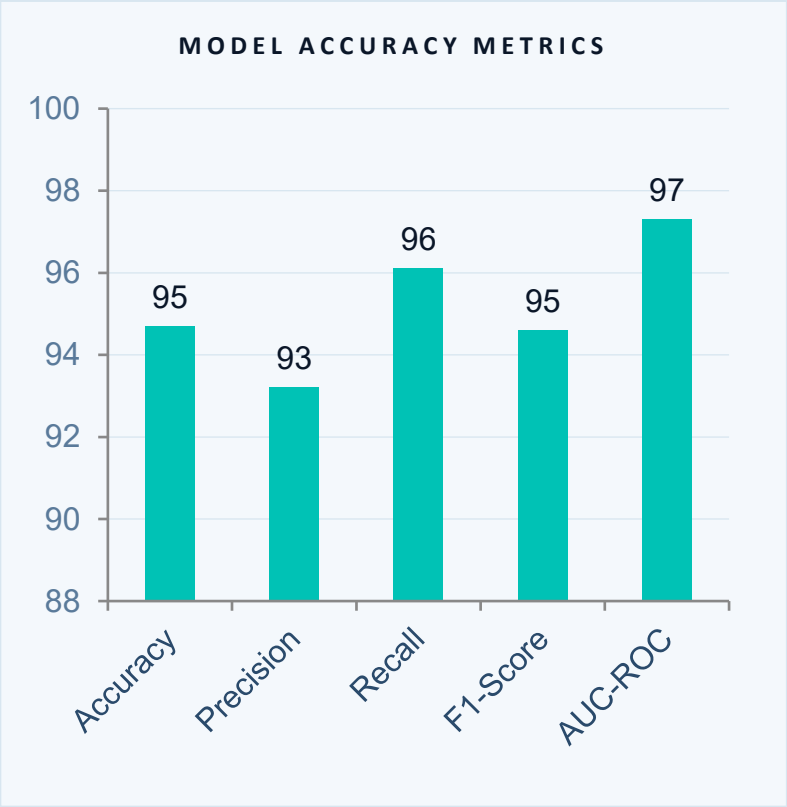
IoT Integration

MQTT + REST API support for 50+ sensor types across industrial equipment families.

MACHINE LEARNING MODEL

MODEL PIPELINE

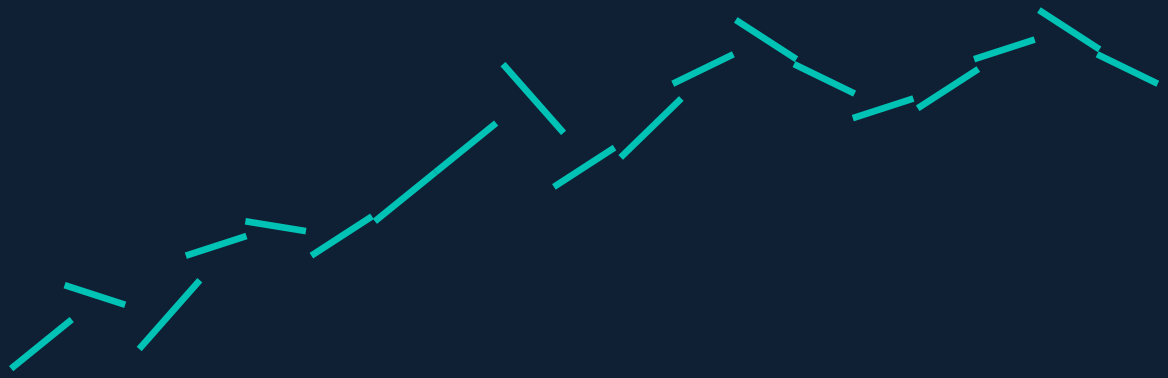
1. **Data Collection** — IoT sensor streams → historical CMMS records
2. **Feature Engineering** — Rolling stats, FFT coefficients, lag features
3. **Ensemble Voting** — RF + XGBoost + LSTM majority vote
4. **Risk Scoring** — 0–100% failure probability per asset



LIVE DASHBOARD — WEB INTERFACE

24	3	1	94.7%
ASSETS	WARNINGS	CRITICAL	ACCURACY

LIVE SENSOR STREAM



PREDICTION OUTPUT

88%

⚠ CRITICAL — Failure likely within 48 hours

Schedule immediate maintenance action

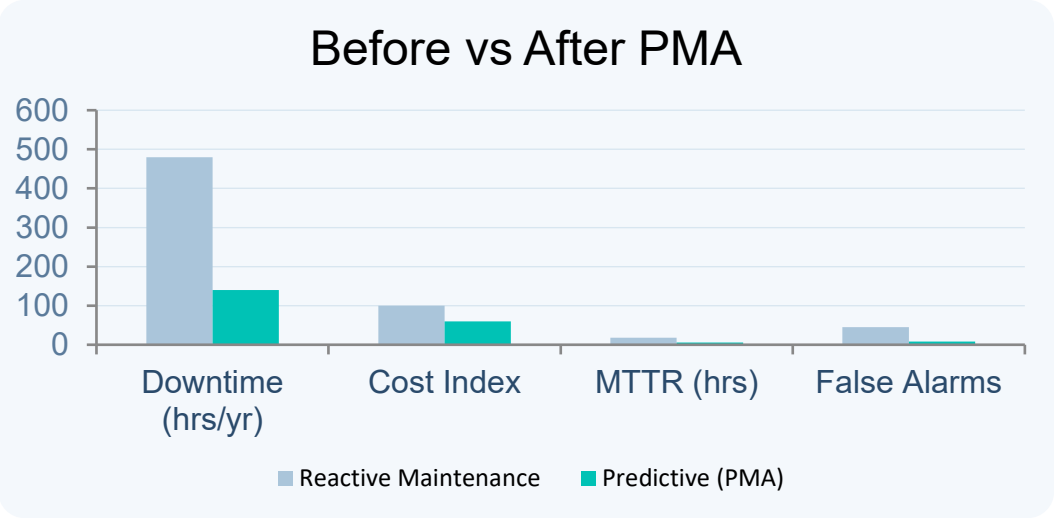
RESULTS & IMPACT

94.7%
Model Accuracy

↓ 35%
Downtime Reduction

↑ 28%
Asset Lifespan

↓ 40%
Maintenance Cost



FUTURE SCOPE

- Digital Twin integration for asset simulation
- Edge AI deployment on Raspberry Pi / Jetson
- Multi-site cloud scalability (AWS IoT Core)
- NLP-powered maintenance report generation
- Reinforcement learning for dynamic scheduling

CONCLUSION

The Predictive Maintenance Agent is Ready.

This project demonstrates how combining real-time IoT sensor data with machine learning models can eliminate reactive maintenance cycles, dramatically cut downtime and costs, and extend asset lifespans across industrial operations.